

Employee Leave Application Documentation

Requirement

Build an employee leave application for our organization.

Architecture

Architecture and Diagram Summary

- **Frontend Layer:** User interface for employees to submit leave requests and for managers to approve them.
- **API Layer:** RESTful API endpoints for handling leave requests and approvals.
- **Application Services Layer:** Business logic for processing leave requests, approvals, and notifications.
- **Persistence Layer:** Data storage for leave requests, employee details, and leave balances.
- **Security Layer:** Authentication and authorization mechanisms to secure the application.
- **External Systems:** Integration with notification services for sending alerts.

Code Design

- `com.example.leave.api`: REST controllers for handling HTTP requests.
- `com.example.leave.service`: Service classes for business logic.
- `com.example.leave.repository`: Data access objects for interacting with the database.
- `com.example.leave.model`: Entity classes representing leave requests and employees.
- `com.example.leave.security`: Security configurations and user details service.

Implementation Plan

Phase 1: Setup

Tasks:

- Set up the development environment.
- Initialize the Spring Boot project.
- Configure PostgreSQL database.

Dependencies: None

Expected Outcome: Development environment ready with database configured.

Phase 2: Database

Tasks:

- Design the database schema.
- Create entity classes.
- Set up Spring Data JPA repositories.

Dependencies: None

Expected Outcome: Database schema designed and entity classes created.

Phase 3: Domain**Tasks:**

- Implement leave request entity.
- Implement employee and manager entities.
- Define leave types and statuses.

Dependencies: Database schema

Expected Outcome: Domain model implemented.

Phase 4: Business Logic**Tasks:**

- Implement leave request service.
- Implement leave approval service.
- Implement leave balance service.

Dependencies: Domain model

Expected Outcome: Business logic implemented.

Phase 5: APIs**Tasks:**

- Create REST controllers for leave requests.
- Create REST controllers for leave approvals.
- Implement request and response DTOs.

Dependencies: Business logic

Expected Outcome: API endpoints implemented.

Phase 6: Security**Tasks:**

- Implement user authentication.
- Define roles and permissions.
- Secure API endpoints.

Dependencies: APIs

Expected Outcome: Application secured with authentication and authorization.

API Design

Endpoints

M e t h o d	Path	P u r p o s e	Request Body	Response Body	S u c c e s s t a t u s	E r r o r s t a t u s
P O S T	/leave-re quests	S u b m i t a n e w l e a v e r e q u e s t	{"employeeId":1, "leave Type": "SICK", "startDat e": "2023-12-01", "endDa te": "2023-12-05"}	{"id":1, "status": "PENDING"}	2 0 1	4 0 0 4 4 5 0 0

M e t h o d	Path	P u r p o s e	Request Body	Response Body	S u c c e s s t a t u s	E r r o r S t a t u s
GET	/leave-requests/{id}	Retrieve available leave request by ID	N/A	{"id":1,"employeeId":1,"leaveType":"SICK","startDate":"2023-12-01","endDate":"2023-12-05","status":"PENDING"}	200	405

M e t h o d	Path	P u r p o s e	Request Body	Response Body	S u c c e s s t a t u s	E r r o r S t a t u s
P A T C H	/leave-requests/{id}/approve	A p p r o v e a l l e a v e r e q u e s t	{"comment": "Approved by manager"}	{"id":1, "status": "APPROVED"}	200	404500

M e t h o d	Path	P u r p o s e	Request Body	Response Body	S u c c e s s t a t u s	E r r o r S t a t u s
P A T C H	/leave-requests/{id}/reject	R e j e c t a l e a v e r e q u e s t	{"comment":"Rejected due to insufficient balance"}	{"id":1,"status":"REJECTED"}	200	404500

M e t h o d	Path	P u r p o s e	Request Body	Response Body	S u c c e s s t a t u s	E r r o r S t a t u s
GET	/employees/{employeeId}/leave-balance	Get available leave balance for an employee	N/A	{"employeeId":1,"leaveType":"SICK","availableLeave":10}	200	40500

Validation and Error Handling

Field Level Validation Rules

- employeeId: Must be a valid employee ID.

- `leaveType`: Must be one of the predefined leave types (SICK, VACATION, etc.).
- `startDate`: Must be a valid date and not in the past.
- `endDate`: Must be a valid date and after the `startDate`.
- `comment`: Must be a non-empty string when approving or rejecting.

Standard Error Response Envelope

```
{
  "timestamp": "2023-10-05T12:34:56.789Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Invalid leave request",
  "details": "The leave request is invalid due to..."
}
```

Business Failures to HTTP Status Codes

Business Failure	HTTP Status Code
Invalid request data	400
Leave request not found	404
Internal server error	500

Data Model

Model

Table	Column	Data Type	Nullabl e	Ke y	Default	Descriptio n
Employee	id	UUID	No	PK	-	Unique identifier for employee
	name	VARCHAR	No	-	-	Employee's name
	email	VARCHAR	No	-	-	Employee's email
	created_at	TIMESTAM P	No	-	CURRENT_TIMESTAMP	Record creation timestamp
	updated_at	TIMESTAM P	No	-	CURRENT_TIMESTAMP	Record last update timestamp

Table	Column	Data Type	Nullabl e	Ke y	Default	Descriptio n
LeaveRequest	id	UUID	No	PK	-	Unique identifier for leave request
	employee_id	UUID	No	FK	-	Employee who requested the leave
	start_date	DATE	No	-	-	Start date of the leave
	end_date	DATE	No	-	-	End date of the leave
	status	VARCHAR	No	-	'PENDING'	Status of the leave request
	created_at	TIMESTAMP	No	-	CURRENT_TIMESTAMP	Record creation timestamp
	updated_at	TIMESTAMP	No	-	CURRENT_TIMESTAMP	Record last update timestamp

ER Diagram

```
{
  "entities": [
    { "name": "Employee", "fields": ["id", "name", "email"] },
    { "name": "LeaveRequest", "fields": ["id", "employee_id", "start_date", "end_date", "status"] },
  ],
  "relationships": [
    { "from": "Employee", "to": "LeaveRequest", "label": "1:N" }
  ]
}
```

Schema Script

```

CREATE TABLE Employee (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  name VARCHAR NOT NULL,
  email VARCHAR NOT NULL UNIQUE,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE LeaveRequest (
  id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
  employee_id UUID NOT NULL REFERENCES Employee(id),
  start_date DATE NOT NULL,
  end_date DATE NOT NULL,
  status VARCHAR NOT NULL DEFAULT 'PENDING',
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

```

Constraints and Indexes

- **Primary Keys:** Employee(id), LeaveRequest(id).
- **Foreign Keys:** LeaveRequest(employee_id) references Employee(id).
- **Unique Constraints:** Employee(email).
- **Check Constraints:** LeaveRequest(status) should be one of 'PENDING', 'APPROVED', 'REJECTED'.
- **Indexes:**
 - CREATE INDEX idx_employee_email ON Employee(email);
 - CREATE INDEX idx_leaverequest_employee_id ON LeaveRequest(employee_id);
 - CREATE INDEX idx_leaverequest_status ON LeaveRequest(status);
 - Reason: To improve the performance of queries filtering by email, employee ID, and leave request status.

User Guide

Overview

This application is designed for employees and managers within our organization to manage leave requests. Employees can submit leave requests, while managers can approve or reject these requests. The application ensures that leave requests are processed efficiently and that employees have an accurate view of their leave balances.

Getting Started

1. **Login:** Employees and managers can log in using their credentials.

2. **Submit Leave Request:** Employees navigate to the leave request section and fill out the necessary details.

3. **Verify Success:** Upon successful submission, employees should see a confirmation message with the status "PENDING".

User Roles and Permissions

Employee

- **Capabilities:** Submit leave requests, view leave balance.
- **Typical workflows:** Submit a leave request, view pending leave requests.
- **Restrictions:** Cannot approve or reject leave requests.

Manager

- **Capabilities:** Approve or reject leave requests, view all leave requests.
- **Typical workflows:** Review a leave request, approve or reject it.
- **Restrictions:** Cannot modify leave balance.

Admin

- **Capabilities:** Manage user accounts, view all leave requests and balances.
- **Typical workflows:** Create new employee accounts, reset passwords.
- **Restrictions:** None.

Core Workflows

Submit Leave Request

This workflow allows employees to request leave. Use this when you need time off for personal reasons.

Steps:

1. Navigate to the "Leave Requests" section by clicking on the menu.
2. Click the "Submit Leave Request" button to open the form.
3. Enter fields with values like:
 - `employeeId: 1`
 - `leaveType: SICK`
 - `startDate: 2023-12-01`
 - `endDate: 2023-12-05`
4. Click the "Submit" button to submit the request.
5. **Expected outcome:** A confirmation message with the status "PENDING".

Common mistakes:

- **Missing** employeeId: Ensure you enter a valid employee ID.
- **Invalid** leaveType: Use one of the predefined leave types (SICK, VACATION, etc.).

Approve Leave Request

This workflow allows managers to approve leave requests. Use this when you need to approve a leave request submitted by an employee.

Steps:

1. Navigate to the "Leave Requests" section by clicking on the menu.
2. Click on the leave request you want to approve.
3. Click the "Approve" button to open the approval dialog.
4. Enter a comment like "Approved by manager".
5. Click the "Approve" button to submit.
6. **Expected outcome:** The leave request status updates to "APPROVED".

Tips:

- Use clear and concise comments for better tracking.
- Ensure the leave dates do not conflict with other approved leaves.

Reject Leave Request

This workflow allows managers to reject leave requests. Use this when you need to reject a leave request submitted by an employee.

Steps:

1. Navigate to the "Leave Requests" section by clicking on the menu.
2. Click on the leave request you want to reject.
3. Click the "Reject" button to open the rejection dialog.
4. Enter a comment like "Rejected due to insufficient balance".
5. Click the "Reject" button to submit.
6. **Expected outcome:** The leave request status updates to "REJECTED".

Tips:

- Provide a clear reason for rejection to help employees understand.
- Ensure the rejection is communicated promptly.

Data Management

Creating Records

To create a new leave request, navigate to the "Leave Requests" section, click "Submit Leave Request", and fill out the required fields. Required fields include employeeId, leaveType, startDate, and endDate.

Editing Records

Editing records is not supported in this application. Once a leave request is submitted, it can only be approved or rejected.

Deleting Records

Deleting records is not supported. Instead, leave requests can be rejected, which effectively denies the request.

Filtering and Search

You can filter and search leave requests by employee ID, leave type, and status using the filters available in the "Leave Requests" section.

Reports and Exports

Reports on leave requests and balances can be generated from the "Reports" section. Exports are available in CSV and PDF formats.

Frequently Asked Questions

Q: How can I check my leave balance?

A: Navigate to the "Leave Balance" section to view your available leave balance.

Q: Can I edit a submitted leave request?

A: No, you cannot edit a submitted leave request. You can only approve or reject it.

Q: Can I <feature question>?

A: No, this feature is not available in the current version.

Troubleshooting

Common Issues